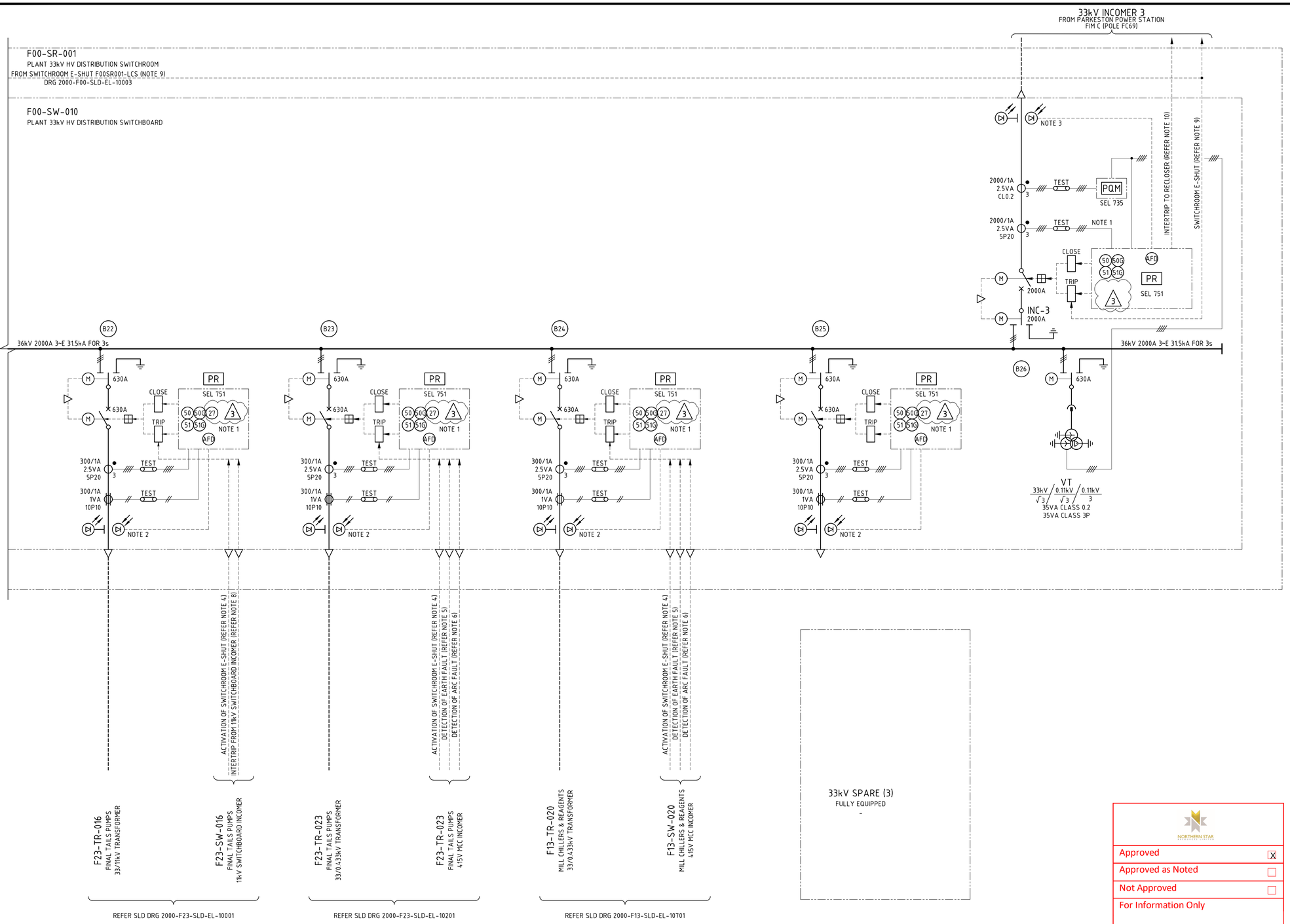


NOTES

1. THE FEEDER PROTECTION RELAY VOLTAGE INPUT SHALL BE WIRED FROM THE CORRESPONDING INCOMER BUS VT.
2. ARC FLASH POINT DETECTORS SHALL BE PLACED IN CABLE ZONES AND CONNECTED DIRECTLY TO THE SEL PROTECTION RELAYS IN EACH PANEL.
3. SWITCHBOARD SHALL BE EXTENSIBLE FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER PER BUS SECTION AS SHOWN.
4. TRANSFORMER 33KV FEEDER SHALL BE TRIPPED ON THE ACTIVATION OF THE SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP THE CIRCUIT BREAKER.
5. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE EARTH FAULT DETECTION OF THE DOWNSTREAM MCC INCOMER ACB.
6. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE ARC FAULT DETECTION OF THE DOWNSTREAM MCC INCOMER ACB.
7. TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS SHALL BE HARDWIRED TO THE TRANSFORMER FEEDER PROTECTION RELAY.
8. THE INTERTRIP OF THE UPSTREAM 33kV FEEDER SHALL BE HARDWIRED FROM THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY EITHER ON A TRANSFORMER TRIP, EARTH FAULT DETECTION OR ARC FAULT DETECTION.
9. THE ACTIVATION OF THE PLANT 33KV DISTRIBUTION SWITCHROOM F00-SR-001 EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP ALL THREE INCOMER CIRCUIT BREAKERS AND THE CORRESPONDING UPSTREAM POLE MOUNTED RECLOSERS.
10. PROTECTION TRIP FROM THE INCOMER CIRCUIT BREAKER SHALL INTERTRIP THE UPSTREAM POLE MOUNTED RECLOSER.

LEGEND

- PR — INCOMER / FEEDER PROTECTION RELAY NOTES 1 & 2
- PQM — POWER QUALITY AND REVENUE METERING
- ARC FLASH SENSOR (POINT TYPE) NOTE 2
- CAPACITIVE VOLTAGE INDICATION
- MECHANICAL OR ELECTRICAL INTERLOCK DEVICE
- 27 — UNDER-VOLTAGE
- 50 — INSTANTANEOUS OVERCURRENT
- 50G 50N — INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
- 51 — TIMED OVERCURRENT
- 51G 51N — TIMED EARTH/NEUTRAL OVERCURRENT
- 64 — EARTH FAULT
- 67 67G — DIRECTIONAL OVERCURRENT/EARTH FAULT
- AFD — ARC FLASH DETECTION

NORTHERN STAR RESOURCES LTD

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

2000-F00-SLD-EL-10006_Rev3

Single line diagram sheet 5

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